# Aryan Singh

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## Education

**University of Waterloo** — Waterloo, ON
*Candidate for Bachelor of Computing and Financial Management (Co-op)* — *Expected Apr. 2030*

- **Relevant Coursework:** Algorithms & Data Structures, Financial Markets, Data Analytics, Financial Reporting

## Technical Skills

**Languages:** Python, Java, C/C++, JavaScript/TypeScript, SQL (MySQL), Bash, HTML/CSS
**Data & Analytics:** pandas, NumPy, NumPy-Financial, Matplotlib, yfinance, scikit-learn
**Frameworks & Technologies:** React, Next.js, Tailwind CSS, Node.js, FastAPI
**Developer Tools & Environments:** Git/GitHub, Linux, VS Code, PyCharm, Jupyter Notebook, Vim, Anaconda

## Experience

**Food & Beverage Service Associate** — Mar. 2023 – Aug. 2025
*Canada's Wonderland* — *Vaughan, ON*

- Executed **300+ high-volume financial transactions** per shift using Oracle POS systems, maintaining 95-99% transaction accuracy and audit-level accountability during peak operating hours.
- Supervised and coordinated day-to-day operations across **5–10 associates per shift** during peak periods, providing task delegation and resolving real-time operational issues to maintain efficient service flow.
- Optimized front-line workflows by prioritizing associate responsibilities and adapting processes in real time, supporting efficient service for **hundreds of guests per shift** in a fast-paced environment.

**International Service Project – Rwanda Missions Trip** — Mar. 2024
*Shelter Them Poverty Relief — Brampton Christian School* — *Kigali, Rwanda*

- Collaborated within a cross-functional team of **15+ members** to plan and execute a community infrastructure project, successfully meeting all project deadlines despite fixed timelines and logistical constraints.
- Managed resources, materials, and on-site execution during the construction of a fully functional cow shelter in partnership with Shelter Them, ensuring on-time completion within a **2-week timeline** under limited resources.
- Planned and executed fundraising and budget allocation initiatives, raising **$2,000+** and coordinating financial planning to ensure on-time completion under limited funding resources.

## Projects

**Robo–Advising Portfolio Optimizer** | *Python, pandas, NumPy, yfinance* — GitHub

- Built an end-to-end portfolio construction pipeline using **pandas and NumPy** to filter equities by currency, liquidity, and market capitalization, with data validation to remove invalid or delisted tickers and handle cross-listed stocks.
- Calculated **annualized volatility** to identify and remove high-risk equities within each sector, controlling portfolio risk while preserving sector-level diversification.
- Constructed a **$1,000,000 CAD market-meet portfolio** by selecting stocks correlated to the S&P 500 and TSX Composite and converting portfolio weights into shares while accounting for FX rates and transaction fees.

**Market Stress & Crisis Simulator** | *Next.js, TypeScript, React, Tailwind CSS, Recharts* — GitHub

- Engineered a portfolio stress-testing platform that replays **3 historical market crises** (2008 GFC, COVID-19 Crash, 2022 Rate Shock), enabling scenario-based analysis of diversified U.S. and Canadian portfolios.
- Built a **portfolio simulation engine** supporting asset weighting, normalization, and configurable rebalancing to model portfolio behavior under extreme market stress rather than return optimization.
- Implemented an analytics pipeline and data processing layer computing **6+ risk metrics**, including maximum drawdown, volatility, Sharpe ratio, and time-to-recovery, to evaluate tail-risk exposure and portfolio recovery dynamics across 12–36 month horizons.

**Equity Trend Analyzer** | *Python, pandas, NumPy, yfinance, Matplotlib, Streamlit* — GitHub

- Developed a **Streamlit** application using historical market data (yfinance) for analysis of equities and computation of performance and risk metrics, including **maximum drawdown**, total return, and annualized volatility.
- Implemented trend detection logic using **linear regression** on log-price data to classify equities as uptrend, downtrend, or no clear trend.
- Built signal analysis tools, including moving average crossovers and **RSI (14)** momentum indicators, with input validation, dynamic plotting, and CSV export to support investment analysis.